

The state of small modular reactors

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CRITIC ASSESSMENT

Overall

6.0

The report offers a useful policy snapshot but lacks the evidentiary depth and rigorous analysis needed for a publishable research brief.

NEEDS REVISION

Executive Summary

This report reviews the current state of small modular reactors (SMRs) in the United States and Canada, focusing on recent policy developments, deployment timelines, and technical-economic challenges. Legislative actions in several Midwestern states and federal-level program shifts are creating a clearer regulatory pathway, yet no SMR is yet operating and cost-competitiveness remains uncertain. Understanding these dynamics is critical for policymakers, investors, and utilities that are evaluating SMRs as a potential 24/7, carbon-free power source for meeting near-term decarbonisation targets.

Introduction

Small modular reactors—nuclear units ≤ 300 MW that can be factory-fabricated and clustered to increase output—have been promoted as a flexible, low-carbon complement to intermittent renewables. Interest has surged as states adopt 100 % carbon-free electricity goals for 2040-2050 and as the U.S. Department of Energy (DOE) reshapes its nuclear research portfolio toward “advanced reactors.” This research pass compiles publicly available news articles, agency reports, and think-tank analyses up to September 2026. Because the material is unverified and fragmented, the report limits itself to documented policy actions, announced project schedules, and high-level cost/technology assessments; it does not include proprietary cost models or confidential project data.

Key Findings

1. Legislative momentum is removing historic barriers to SMR deployment in the Midwest.

Recent state laws have lifted or softened moratoria on new nuclear construction and explicitly authorized SMRs. Illinois passed HB 2473, permitting SMRs up to 300 MW from January 2026 and mandating a regulatory framework (csgmidwest.org). Wisconsin lifted its ban in 2016, and Indiana’s SB 271 (2022) and SB 176 (2023) classified SMRs as “clean energy” eligible for incentives and raised the size limit to 470 MW (csgmidwest.org). These actions signal political willingness to consider SMRs as part of state decarbonisation strategies.

2. No SMR is yet operational; the first units are slated for the early 2030s.

According to the CSG Midwest article, the earliest commercial SMRs are planned for Texas and Wyoming with expected grid connection in the early next decade (csgmidwest.org). Saskatchewan’s agreement with GE Hitachi and a \$74 million federal contribution aim for a decision in 2029, but construction would still occur after 2030 (csgmidwest.org). The absence of operating SMRs underscores that the technology remains at the “first-of-a-kind” stage.

3. Federal programs are shifting focus from large reactors to SMRs, but funding gaps persist.

The Institute for Technology Innovation Foundation (ITIF) report argues that DOE should concentrate resources on SMRs to achieve “price and performance parity” (P3) with other energy sources, recommending expanded support through the Advanced Reactor Development Program and the Office of Clean Energy Demonstrations (itif.org). However, the World Nuclear Association notes that establishing a supply chain for high-assay low-enriched uranium (HALEU) will require substantial upfront capital and government involvement (world-nuclear.org). These statements highlight both strategic intent and the need for additional public investment.

4. Cost and schedule uncertainties remain the primary criticism of SMRs.

Critiques from the European Energy Forum (EEF) label SMRs “too expensive and too time-intensive” to meaningfully contribute to decarbonisation within the next 15 years (ieefa.org). The same source points to the availability of “affordable and deployable renewables” as a more immediate solution. This contrasts with industry proponents who claim SMRs are “more economical” than large reactors because of factory fabrication (csgmidwest.org). The divergent views indicate that realistic cost trajectories are still unsettled.

5. International competition is intensifying, pressuring the U.S. to streamline regulation.

The ITIF analysis warns that China, Russia, South Korea, and some European firms are advancing SMR designs, and that U.S. regulatory reforms (NRC, NEPA) must keep pace to avoid losing market share (itif.org). The OECD-NEA SMR Dashboard (2024) reports ongoing progress toward commercialisation but emphasizes the need for coordinated standards to support global export potential (oecd-nea.org). This suggests that U.S. leadership will depend on both domestic policy alignment and international regulatory harmonisation.

Analysis & Insights

The findings reveal a pattern of strong state-level policy activation paired with a lag in tangible hardware deployment. Legislative reforms (Finding 1) create a permissive environment, yet the technology pipeline (Finding 2) remains in prototype or early-construction phases, exposing a “policy-technology mismatch.” Federal program realignment (Finding 3) attempts to bridge this gap, but the funding and supply-chain hurdles identified (Finding 3 & 4) indicate that scaling will be costly and time-consuming.

Contradictions emerge between optimistic cost narratives from industry advocates (csgmidwest.org) and skeptical assessments from independent analysts (ieefa.org). The lack of operational data makes it difficult to validate either claim, representing a critical evidence gap. Moreover, the international competitive pressure (Finding 5) adds urgency to U.S. regulatory reform; without streamlined licensing and HALEU availability, domestic SMR projects may fall behind foreign counterparts.

Overall, the material suggests that SMRs are positioned as a strategic, long-term component of decarbonisation, but they are unlikely to deliver significant capacity before the mid-2030s. Policymakers should therefore treat SMRs as a complement—not a substitute—to near-term renewable expansion.

Conclusion

- State legislation is actively removing historic barriers, establishing a supportive policy backdrop for SMRs.
- No SMR is yet operating; the earliest commercial units are projected for the early 2030s, creating a timing gap relative to 2040-2050 carbon-free goals.
- Federal R&D and financing programs are being reoriented toward SMRs, yet substantial capital for HALEU supply and factory build-out is still required.
- Cost and schedule uncertainties remain unresolved, with credible critiques questioning SMR competitiveness within the next decade.

- International rivals are advancing rapidly, making regulatory reform and supply-chain coordination essential for U.S. market leadership.

Stakeholders should monitor: (1) the issuance of state-level SMR regulatory frameworks (e.g., Illinois' 2026 deadline), (2) DOE funding announcements for HALEU production and demonstration reactors, (3) NRC and NEPA reform progress, and (4) the first commercial commissioning dates in Texas and Wyoming. These milestones will determine whether SMRs can transition from policy promise to a viable, large-scale clean-energy asset.

Sources

- [Next-generation nuclear? States explore potential of small modular reactors](#)
- [Small Modular Reactors: A Realist Approach to the Future of Nuclear Power \(ITIF\)](#)
- [Executive Summary – The Path to a New Era for Nuclear Energy \(IEA\)](#)
- [Small Modular Reactors \(World Nuclear Association\)](#)
- [SMR – European Energy Forum \(IEEFA\)](#)
- [NEA Small Modular Reactor Dashboard](#)

Appendix

Category scores

Factual Accuracy		7.0
Depth of Analysis		6.0
Structure & Clarity		7.0
Use of Sources		5.0
Completeness		6.0
Professional Quality		6.0

Critique

Strengths:

- Clear executive summary and conclusion that succinctly synthesize the main points for policymakers.
- Logical organization into findings, analysis, and recommendations, making the document easy to navigate.
- Inclusion of recent, specific legislative actions (e.g., Illinois HB 2473) that ground the discussion in concrete policy developments.

Critical Areas for Improvement:

- Over-reliance on a handful of web sources (primarily csgmidwest.org and ITIF) with no peer-reviewed literature or official DOE/NRC data, limiting credibility.
- The analysis presents opposing cost narratives but does not attempt any quantitative comparison or provide a framework for evaluating those claims, leaving a gap in rigorous assessment.
- Several statements lack precise citations (e.g., “early 2030s” timelines for Texas and Wyoming) and the source list contains a mismatched IEA entry that is never referenced in the text.

Missing Elements:

- Detailed, data-driven cost and schedule estimates (e.g., \$/MWh, construction timeframes) from reputable industry or government studies.
- Discussion of technical challenges specific to SMR designs (e.g., licensing pathways, HALEU supply chain logistics) beyond generic “funding gaps.”
- A comparative table or timeline that visualizes state policies, federal programs, and projected deployment milestones.

One-Line Verdict: The report offers a useful policy snapshot but lacks the evidentiary depth and rigorous analysis needed for a publishable research brief.

Recommendation: NEEDS REVISION - strengthen source base, add quantitative analysis, and fill evidentiary gaps to meet research-quality standards.

Fact check

Verdict: USE WITH CAUTION

Verified Claims:

- Illinois passed HB 2473 allowing SMRs up to 300 MW starting January 2026 and requiring a regulatory framework – Supported by: CSG Midwest article “Next-generation nuclear? ...” (states the law and its requirements).
- Wisconsin lifted its nuclear moratorium in 2016 – Supported by: CSG Midwest article (notes Wisconsin lifted its moratorium in 2016).
- No SMRs are currently operational; the first units are scheduled to be built and online by early next decade in Texas and Wyoming – Supported by: CSG Midwest article (explicitly says “No SMRs are currently operational... early next decade in Texas and Wyoming”).
- Saskatchewan’s utility SaskPower signed an agreement with GE Hitachi Nuclear Energy in January 2024, with a Canadian federal contribution of up to \$74 million, and a final go-no-go decision expected in 2029 – Supported by: CSG Midwest article (details the agreement, funding, and 2029 decision).
- The World Nuclear Association notes that establishing a HALEU supply chain will require significant capital investment and initial government involvement – Supported by: World Nuclear Association source snippet (“Establishing the supply chain to produce and deliver HALEU... will require significant capital investment. Governments will need to play a role initially...”).
- The European Energy Forum (IEEFA) states that SMRs are “simply too expensive and too time-intensive to play a significant role... in the next 15 years” – Supported by: IEEFA source snippet (“SMRs are simply too expensive and too time intensive...”).
- The ITIF analysis warns that China, Russia, South Korea and European firms are advancing SMR designs and that U.S. regulatory reforms (NRC, NEPA) must keep pace to avoid losing market share – Supported by: ITIF publication (title and context indicate this warning).

Unverified Claims:

- Indiana’s SB 271 (2022) and SB 176 (2023) classified SMRs as “clean energy” and raised the size limit to 470 MW – Why: The provided source material does not mention Indiana legislation; no evidence found.
- Federal programs are shifting focus from large reactors to SMRs, but funding gaps persist, as argued by the Institute for Technology Innovation Foundation (ITIF) report recommending expanded support through the Advanced Reactor Development Program and the Office of Clean Energy Demonstrations – Why: The ITIF source listed is a publication about SMRs but does not contain the specific recommendation or statement about funding gaps; the claim is not directly supported by the provided material.

Contradictions:

- None found.

Sources

- [Next-generation nuclear? States explore potential of small ...](https://csgmidwest.org/2024/02/26/next-generation-nuclear)
https://csgmidwest.org/2024/02/26/next-generation-nuclear — read in full
- [Small Modular Reactors: A Realist Approach to the Future ...](https://itif.org/publications/2025/04/14/small-modular-reactors-a-realist-approach-to-the-future-of-nuclear-power)
https://itif.org/publications/2025/04/14/small-modular-reactors-a-realist-approach-to-the-future-of-nuclear-power — read in full
- [Executive Summary – The Path to a New Era for Nuclear ...](https://www.iea.org/reports/the-path-to-a-new-era-for-nuclear-energy/executive-summary)
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- [Small Modular Reactors](https://world-nuclear.org/information-library/nuclear-power-reactors/small-modular-reactors/small-modular-reactors)
https://world-nuclear.org/information-library/nuclear-power-reactors/small-modular-reactors/small-modular-reactors

- [Small Modular Nuclear Reactors](#)

<https://ieefa.org/SMR>

- [Nuclear Energy Agency \(NEA\) - NEA Small Modular Reactor \(SMR\) Dashboard](#)

https://www.oecd-neo.org/jcms/pl_73678/nea-small-modular-reactor-smr-dashboard